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MILITARY SPECIFICATION

TEST SET MARK 127 MOD 4

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

L SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of Test Set Mark 127 Mod 4, Naval Ordnance Systems Command LD 293403.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Comer
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4925

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MILITARY

MIL-B-131	Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat-sealable

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-731	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111 -5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications for-m a part of this document to the extent specifiee herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND (CAGE Code 10001)

LD 284866	Preparation for Delivery for Test Set Cables
LD 293403	Test set Mark 127 Mod 4

(Applications for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station, Attn: Code 802 Louisville, KY 40214-5001.)

2.2 Order of precedence In the event of conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.1.

3.1.2 Fabrication. The Test Set Mark 127 Mod 4 shall be manufactured in accordance with LD 293403 and all documents as specified herein. Inspection and acceptance shall be in accordance with section 4.

3.2 Data No data are required by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

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3.3 Performance requirements and product characteristics. The Test Set Mark 127 Mod 4 shall meet the following performance requirements and product characteristics.

3.3.1 Calibration requirements.

3.3.1.1 Power supply circuit calibration. After the power supply circuit has been calibrated in accordance with 3.5.1, there shall be a potential of 510 ± 10 volts between the positive (+) jack P6, and the negative (-) jack P7, when the test set is connected to a 115 volt AC supply and the ON-OFF switch is at the ON position for at least 15 minutes.

3.3.1.2 Time delay circuits calibration.

33.1.2.1 Five-second timing delay circuit calibration. After the five-second timing delay circuit has been calibrated in accordance with 3.5.2.1, the five-second timing delay shall operate 5 ± 0.25 seconds after the START switch is depressed.

3.3.1.2.2 One-second timing delay circuit calibration. After the one-second timing delay circuit has been calibrated in accordance with 3.5.2.2 the one-second timing delay shall operate 1 ± 0.1 second after the START switch is depressed

3.3.1.3 VOLTAGE AND LEAKAGE meter calibration. After the VOLTAGE AND LEAKAGE meter has been calibrated in accordance with 3.5.3, the test shall successfully meet all requirements of 3.3.2.1 and 3.3.2.1

3.3.2.1 Battery insulation resistance After the VOLTAGE AND LEAKAGE meter has been calibrated in accordance with 3.5.3, the test set shall measure battery insulation resistance with an accuracy of ± 5 percent.

3.3.2.2 Battery voltage. After the VOLTAGE AND LEAKAGE meter has been calibrated in accordance with 3.5.3, the test set shall measure no-load and loaded battery voltages with an accuracy of ± 3 percent.

3.3.2.3 Battery short circuit currents. The test set shall measure battery short circuit currents with an accuracy of ± 3 percent.

3.3.3 Continuity.

3.3.3.1 Cable assemblies CA-731 and CA-802. The D.C. resistance of each circuit of Cable Assemblies CA-731 and CA-802 shall not exceed 0.15 ohm. The continuity of Cable Assembly CA-731 shall be regarded as acceptable when all tests of 4.5.2 requiring the test set to be energized are successfully performed.

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3.3.3.2 Continuity between pin V of adapter cables connector PI and pin N of adapter cables connector P2. The D.C. resistance between pin V of the 19-pin connector, P1, and the 14-pin connector, P2, shall not exceed 0.15 ohm.

3.3.3.3 Continuity between pin B of test stand connector and ground The resistance between pin B of the Test Stand connector and ground shall not exceed 0.15 ohm, when the TEST switch is not operated

3.3.3.4 VOLTAGE AND LEAKAGE meter circuit, meter by passed. With a jumper connected between the terminals of the VOLTAGE AND LEAKAGE meter, the test set energized and the TEST SELECTOR switch at position 2, the test set shall successfully meet the requirements of 3.3.3.4.1 and 3.3.3.4.2.

3.3.3.4.1 Continuity between pins A and C of ADAPTER CABLES connector, P3. With the NO LOAD-LOAD switch at the LOAD position, the resistance between pin A and pin C of the 3-pin ADAPTER CABLES connector, P3, shall remain 0.15 ohm or less for about 5 minutes (see 33.1.21) after the START switch is momentarily depressed.

3.3.3.4.2 Continuity between pins B and C of adapter cables connector, P3. The resistance between pin B and pin C of the 3-pin ADAPTER CABLES connector, P3, shall remain 0.30 ohm or less for about 5 minutes (see 3.3.1.21) after the start switch is momentarily depressed.

3.3.3.5 Continuity through self controlled rotary switch, REL-X. With the test set energized and the TEST SELECTOR switch set at 1, the resistance between pin V of ADAPTER CABLES connector PI and each contact listed by table IV (see 4.5.2.6.5) for both positions of the NO LOAD-LOAD switch shall not exceed 0.15 ohm.

3.3.3.6 Continuity through the ON-OFF switch.

3.3.3.6.1 ON position. When the ON-OFF switch is at the ON position the red pilot lamp PL 15 shall light, remain LIT for the entire period the ON-OFF switch is at the ON position. The ON-OFF switch shall not be regarded as defective on the basis of PILOT Lamp PL 15 not lighting when replacement of the pilot lamp and/or fuse corrects this deficiency.

3.3.3.6.2 OFF position. When the ON-OFF switch is at the OFF position, the VOLTAGE and LEAKAGE Meter shall be effectively by-passed and Pilot Lamp PL 15 shall not be lit.

3.3.3.7 D.C resistances.

3.3.3.7.1 VOLTAGE and LEAKAGE meter coil resistance. The resistance of the coil of the VOLTAGE and LEAKAGE meter shall be as specified by Drawing 1402298.

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3.3.3.7.2 Resistor R14. With the Panel-Chassis Assembly, Drawing 1362835 removed from the test set case and not energized, the TEST SELECTOR switch at the OFF position and the 5 Meg-50 Meg switch at the 50 Meg position the resistance between the chassis and the negative (-) test jack, P7, shall be 0.1 megohm ± 1 percent.

3.3.3.7.3 Resistor R15. With the Panel-Chassis Assembly removed from the test set ease and not energized, the TEST SELECTOR switch at the OFF position and the 5 Meg-50 Meg switch at the 5 Meg position the resistance between the chassis and the negative (-) test jack P7, shall be 10,000 ohms ± 1 percent

3.3.4 Insulation resistance.

33.4.1 Conditioning. The requirements of 3.3.4.2 and 33.4.3 shall be successfully met at a relative humidity of 60 percent or more within one-half hour after the test set has been conditioned for at least 24 hours at a temperature of 77 ± 2°F and a relative humidity of 75 ± 5 percent.

3.3.4.2 500 megohms. With the wipers of the self controlled rotary switch, REL-X at the HOME position, the TEST SELECTOR at the OFF position, the NO-LOAD-LOAD switch at the NO LOAD position and Cable Assembly CA-731 disconnected from the power source the resistance between each pair of points listed by table V (see 4.5.2.7.2g) shall be 500 megohms or more, when 500 ± 50 volts D.C. is applied for 60 ± 5 seconds.

33.43 50 Megohms. With the wipers of RE-X and the test set switches at the positions specified by 3.3.4.2 and with Cable Assembly CA-731 disconnected from the power source, the resistance between each pair of points listed by table VI (see 4.5.2.7.2h) shall be 50 megohms or more when 500 ± 50 volts D.C. is applied for 60 ± 5 seconds.

3.3.4.4 Cable assembly CA-802. The insulation resistance of Cable Assembly CA-802 shall be as specified on Drawing 893800.

3.3.5 Operation of the self controlled rotary switch REL-X The self controlled switch REL-X shall operate as required by the applicable quality conformance tests of 4.5.2. Operation of the self controlled rotary switch is indicated by the HOME lamp and pilot lamp PL 1 through PL 11 lighting when required.

3.3.6 Continuity test switch operation. Pilot Lamp PL 14 shall light and be extinguished as required when performing the tests of 4.5.2.8

3.3.7 Test circuits operation. Pilot Lamp PL 12 shall light and be extinguished as required when performing the test of 4-5.2.9.

3.3.8 Fuse F1. There shall be continuity through Fuse F1 as evidenced by pilot lamp PL 15 lighting when the ON-OFF switch is set at ON and remaining lit for the entire period the ON-OFF switch remains at ON (see 3.3.3.6.1).

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3.4 Durability requirements.

3.4.1 Repeated package impact. After being mounted and vibrated as specified by 4.6.1, the test set shall successfully meet the requirements of all subsequent tests required by figure 1.

3.4..2 Shock, transportation, storage and handling. The cornerwise drop test specified by 4.6.2 shall cause no damage to the exterior shipping container or the case of the test set. There shall be no loosening of hardware or other mechanical damage that may impair the function of the test set. Following the cornerwise drop test, the test set shall successfully meet the requirements of all subsequent tests required by figure 1.

3.4.3 Vibration, transportation, storage and handling. The test set after being mounted and vibrated as specified in 4.6.3 shall show no damage and shall successfully meet the requirements of all quality conformance tests of 4.5.2

3.5 Calibration procedures. Each test set shall be calibrated in accordance with 3.5.1, 3.5.2 and 3.5.3. Calibration shall be performed at a temperature between 65°F and 95°F after the test set has been connected to a 115 volt AC. supply by Cable Assembly CA-731, the ON-OFF switch has been set at ON and the test set allowed to warm up for 15 minutes.

WARNING

Since high voltages are present in the test set power supply,
be extremely careful to avoid touching open leads or components.

3.5.1 Power supply circuit calibration procedure Procedures 3.5.1a through 3.5.1n shall be performed in sequence with the Panel Chassis Assembly, Drawing 1362835, removed from the Instrument Case, LD 276015.

a. Set the ON-OFF switch at OFF.

b. Connect the negative lead of the external vacuum tube voltmeter, Item 4.7a, to the g-rid cap of the 6J7 tube and the positive lead to pin 8 of the 6J7.

c. Adjust RI 7 for approximate mid-rotation.

d. Set the ON-OFF switch to ON and allow one minute for voltages to stabilize.

e. Adjust R18 until external VTVM indicates 10 ± 1 volts. (Note: If adjustment is critical, adjust RI 7 until NE-32 glows.)

f. Set ON-OFF switch to OFF.

g. Remove positive and negative leads of external VTVM.

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h. Connect negative lead of VTVM to negative (-) jack P7 located on the chassis and positive lead to the center tap of R17.

i. Set ON-OFF switch to ON.

j. Adjust R17 until external VTVM indicates 230 volts.

k Set ON-OFF switch to OFF.

1. Remove positive lead of external VTVM from center tap of RI 7 and connect to positive (+) jack P6 located in the chassis.

m. Set ON-OFF switch to ON, and adjust R18 until external VTVM meter indicates 510 volts.

n. Repeat steps f through m as necessary to achieve 230 ± 10 volts from center tap of R17 and negative (-) jack P7 and 510 ± 10 volts from P6 to P7.

o. Disconnect the external VIVM from the circuit.

3.5.2 Time delay circuit calibration procedures.

3.5.2.1 Five-second timing delay calibration procedure. With the panel-chassis assembly removal from the ease, procedures 3.5.2.1a through 3.5.2.1i shall be performed in sequence.

a. Set the TEST SELECTOR switch at 1.

b. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

c. Set the ON-OFF switch at OFF.

d. Connect a jumper across the terminals of the VOLTAGE and LEAKAGE meter. This jumper must be as short as possible and make positive electric contact with both meter terminals.

e. Connect the external electric timer specified by 4.7b to pins B and C of the 3 contact ADAPTER CABLES connector, P3.

f. Set the ON-OFF switch at ON.

g. Set the TEST SELECTOR switch at 2.

h. Momentarily depress the START switch. The timer shall run for 5.0 ± 0.25 seconds. If the timer runs for a period other than 5.0 ± 0.25 seconds. momentarily depress the START switch again. Adjust the slider on resistor R5 nearest the end of R5 which is connected with a gray wire until the timer stops 5.0 ± 0.25 seconds after the START switch is momentarily depressed.

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i. Remove external electric timer.

3.5.2.2 One-second timing delay circuit calibration. Immediately after performing the procedures of 3.5.21, the procedures of 3.5.2.2a through 3.5.2.2e shall be performed in sequence.

a. Set the TEST SELECTOR switch at 5.

b. Connect the external electric timer specified by 4.7b to the positive (+) and negative (-) binding posts, BP1 and BP2, located on the test set panel.

c. Depress and hold the START switch until the timer stops. The timer shall run for 1.0 ± 0.1 second. If the timer runs for a period other than 1.0 ± 0.1 second, depress the START switch again. Adjust tap of resistor R5 nearest the end of R5 which is connected with a purple wire until the timer runs for 1.0 ± 0.1 second when the START switch is depressed and held

d. Remove the external electric timer.

e. Remove the jumper from the VOLTAGE and LEAKAGE meter.

3.5.3 Voltage and leakage meter calibration procedure. With the panel-chassis assembly removed born the case, procedures 3.5.3a through 3.5.3q shall be performed in sequence.

a. Set the ON-OFF switch to OFF.

b. Adjust the zero adjustment of the VOLTAGE and LEAKAGE meter until the meter indicates zero.

c. Connect the 5-megohm resistor specified by 4.7f from pin B of the TEST STAND connector to pin A of the 19-pin ADAPTER CABLES connector.

d. Set the ON-OFF switch at ON.

e. Set the NO LOAD-LOAD switch at NO LOAD.

f. Set the 5 Meg-50 Meg switch at 5 Meg.

g. Set the TEST SELECTOR switch at 3.

h. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

i. Turn the VTVM ADJ control to obtain an indication of zero on the VOLTAGE and LEAKAGE meter.

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j. Momentarily press the START switch, adjust resistor R2 until a reading of 46 is obtained- Lock the shaft of R2 (NOTE It may become necessary to press the ADVANCE and START switches several times to assure a meter reading of 46, due to the automatic 5-second timing feature)

k. Disconnect the end of the 5-megohm resistor connected to pin B of the TEST STAND connector.

L Connect the 50-megohm resistor specified by 4.7g from pin B of the TEST STAND connector, to pin C of the 3-pin ADAPTER CABLES connector.

m. Set the 5 Meg-50 Meg switch at 50 Meg.

n. Set the TEST SELECTOR switch at position 4.

0. Turn the VTVM ADJ control to obtain a reading of zero on the VOLTAGE and LEAKAGE meter.

p. Momentarily press the START switch. The VOLTAGE and LEAKAGE meter shall indicate between 46 and 50 (46 + 2 scale divisions).

q. Disconnect the resistors.

3.6 Clean-up. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign matter. Burrs and sharp edges as well as resin flash shall be removed.

3.7 Soldering. Electrical connections shall be mechanically secure and electrically continuous before soldering.

3.8 Workmanship. The equipment, including parts and accessories, shall be constructed and finished in compliance with all requirements of this specification. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage. Parts shall be free from burrs and sharp edges. The workmanship standards exhibited in any approved first article sample, shall determine the requirements of the contract or purchase order relative to workmanship insofar as not covered by applicable requirements on drawings, specifications, and standards (see 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproval by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet alI requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

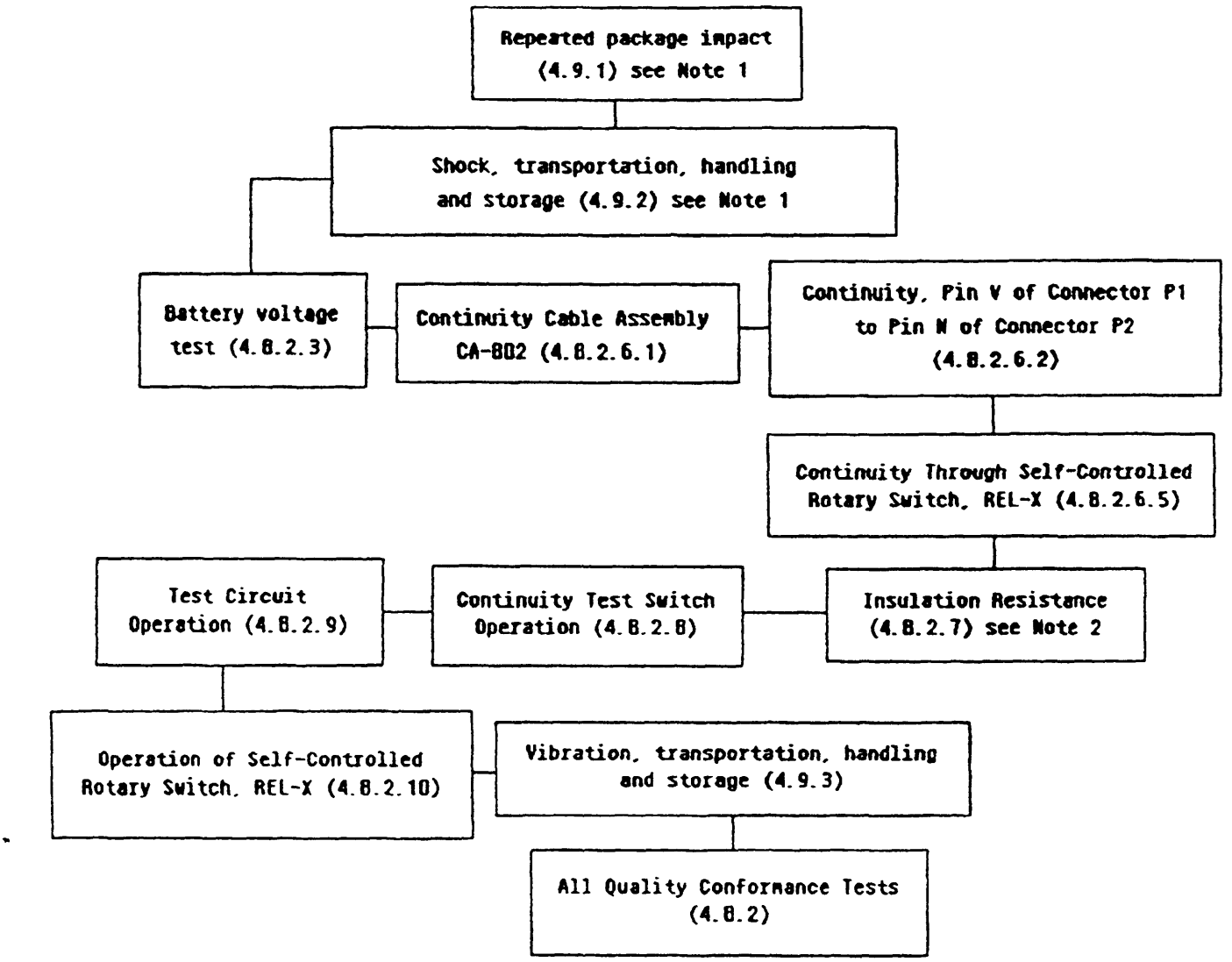
a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
c. Periodic production inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test.

4.4 First article inspection. A first article sample of one test set manufactured in accordance with 3.1 and packaged in accordance with Section 5 (packaging level to be specified by contract or order), shall be subjected to the examination and tests of figure 1 at an activity to be determined by the contracting activity (see 6.2). Durability tests shall be conducted in the sequence shown by the figure. Where indicated by figure 1, inspection and quality conformance tests shall be performed following durability tests. Acceptance of the sample shall be based on no defects in this sample. Further production of test sets by the supplier prior to acceptance of the first article sample shall be at the supplier's risk

4.5 Quality conformance inspection. Each test set shall be subjected to all inspections and tests of 4.8. Any test set that does not successfully meet all requirements when subjected to these inspections and tests shall be regarded defective.

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1. When the purchase document specifies Level C packaging and packing. this test is not required; other - preceding the vibration, transportation, handling and storage tests are required only for the first article sample.

2 Prior to performing the vibration. transportation. handling and storage test, insulation resistances shall be measured wthout removing the panel-chassis assembly from the case. Contacts to GND listed by tables V and VI shall be made to pin B of the TEST STAND connector. The following insulation resistances cannot be measured at this time:

a. Wiper REL-X4 to Ground
b. Wiper REL-XS to Ground
c. Moving Contact REL-IC to Ground
d. Moving Contact REL-ID to Ground

3 All insulation resistance shall be measured after vibration, transportation, handling and storage.

FIGURE 1. First article or periodic production testS-

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4.6 Periodic sampling tests. After successful completion of the quality conformance inspection of 4.8, the tests of figure 1 shall be conducted on a random sample of one test set selected by the Government from each lot of test sets offered for acceptance, unless otherwise specified (see 6.2). Unless otherwise specified in the purchase document, a lot shall be one month's scheduled production or 25 test sets, whichever is less. When a unit fails to pass any periodic sampLing test or specified subsequent quality conformance tests, examination of the unit shall be made by the contractor to analyze the cause of the failure and corrective action to be taken. When cause of the failure is determined to be supplier nonconformance to the quality conformance test requirements of this specification or associated drawings, the contractor may be required to screen the lot and repair or replace the discrepant part or material and resubmit the lot for complete retesting.

4.7 Test equipment The following items of test equipment are required to perform the quality conformance tests set forth in this specification.

a. Vacuum tube voltmeter, minimum input impedance of 10 megohms, ± 3 percent accuracy.

b. Electric timer, range 0-10 seconds, accuracy ± 0.01 second.

c. DC regulated power supply, 1 volt ± 1 percent.

d. Dry cell, 6-volt, heavy duty.

e. Resistor, O.1-megohm, accuracy ± 10 percent, 2 Watt.

f. Resistor, 5.O-megohm ± 1 percent.

g. Resistor, 50.0-megohm ± 1 percent.

h. Ammeter, O - 100 amperes DC, accuracy ± 1 percent.

i. Precision ohmmeter capable of measuring 0.15 ohm and 0.30 ohm with a ± 0.1 percent accuracy.

j. Resistor, 50,000-ohm ± 10 percent.

k. Potentiometer, 50,000 ± 20 percent.

L Decade resistor, 1,111 ohms total, in steps of 0.1 ohm, accuracy ± 0.1 percent.

m. A precision ohmmeter or resistance bridge capable of measuring 10,000 ohms and 0.1 megohm with a ± 0.1 percent accuracy.

n. A resistance measuring device with a test potential of 500 ± 50 volts D. C., capable of measuring 500 megohms or more with a ± 5.0 percent accuracy.

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4.8 Examination and tests.

4.8.1 Examination. Each test set shall be visually and dimensionally examined in accordance with table I to determine compliance with this specification and its associated drawings as specified in 3.1.2. Examination in accordance with table I shall not relieve the contractor of his responsibility to meet all drawing requirements not listed by the table Unless otherwise specified (see 6.2), the classification for all nonconforming unlisted characteristics and workmanship defects shall be determined by the contracting officer. A test set is defective if any of the requirements listed in table I are not met.

TABLE I. Classification of visual and dimensional requirements.

Classification	Requirement
CRITICAL	
None	
MAJOR	
101	Descriptive marking on top and front of instrument case shall be legible and correct (Ref 1379022).
102	Name plate shall be legible and correct (Ref 1362835).
103	Panel marking shall be correct, legible and properly positioned (Ref 1362835).
104	Deleted by Amendment 1
105	Paint shall cover entire surface of instrument case except rubber seal and hardware.
106	Paint shall be free of chips, blisters and deep scratches.
107	Cable Assemblies CA-731 and CA-802 shall be stowed in the cable well of the test set (Ref 1379022).
108	Hinges 1854075, two hinges, and Drawbolts 1854076, two drawbolts, shall hold the lid of the instrument case to the bottom with positive action (Ref 491048, 491049).

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109	A synthetic rubber strip shall be cemented to the angle (Ref 491049).
110	3.000 ±.005 MIN (Ref 4862%)
111	1.000 ±.015 MIN (Ref 4%2%)
*112	Component identification shall be correct, legible and properly positioned (Ref 893786, 893787, 1261853, 1362835, 1402300).
*113	All parts and components shall be free of resin flash, spattered or excess solder, metal chips and other foreign material (Ref 3.6).
l 114	Burrs and sharp edges shall be removed from all parts and components (Ref 3.6).
*115	Electrical connections shall be mechanically secure before soldering.
**116	Welding shall be complete, free of sharp edges and protrusions and indicative of good workmanship (Ref 491047, 491049, 491050, 893792, 1125184).

MINOR

201	Markings on top and front of instrument case shall be centrally located (Ref 1379022).
202	Identification marking of Cable Assemblies CA-731 and CA-802 shall be legible and correct (Ref 490792, 497136).
203	Rubber tape shall be wrapped around CA-802 Cable Subassembly 893800 to obtain effective clamping (Ref 490792, 893800).
204	25.125 ± 0.020 (Ref 491047)
205	10.062 ± 0.020 (Ref 491047)
206	2.500 ± 0.020 (Ref 491047)
207	14.187 ± 0.030 (Ref 491049)
208	1.468 ± 0.010 (Ref 491049)
209	25.125 ± 0.030 (Ref. 491050)
210	10.062 ± 0.030 (Ref 491050)

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211	11.437 ± 0.030 (Ref 491050)
212	2.000 ± 0.030 (Ref 491050)
213	3.625 ± 0.060, four places (Ref 491050)
214	72.000 + 1.000 - 0.000 (Ref 497136)
215	24.000 + 1.000 -0.000 (Ref 893800)
216	Jewels of Pilot Lamps shall be the following colors:

PL 1 through PL 11, PL 13	Amber
PL 12, PL 14	Green
PL 15	Red

1. All dimensions are in inches.

2 Requirements identified with an asterisk (“) must be verified prior to assembly of the test set.

3. Welding requirements identified with a double asterisk (**) must be verified prior to painting the Instrument Case (LD 276015).

4. Paragraph references apply to paragraphs of this specification.

5. For other units of inspection which are components of the unit covered by this table usc Test Set Mk 127 Mod 4 Assembly Drawing 1379022

4.8.2 Tests.

WARNING

Since high voltages are present in the test set power supply, be extremely careful to avoid touching open leads or components. Wherever practical, tests should be conducted with the panel-chassis asssembly (Drawing 1362835) installed in the instrument case LD276015. Before removing the pane-chassis assembly from the case, disconect the test set from the power source.

4.8.2.1 Power supply circuit operation. Procedures 4.8.21a through 4.8.2.1e shall be performed in sequence at a temperature between 65°F and 95°F after the power supply has been calibrated in accordance with 3.5.1. The panel-chassis assembly shall be removed from the case, energized (see 6.4.7) and allowed to warm up for 15 minutes before beginning the test.

a. Set the ON-OFF switch at OFF

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b. Connect the positive terminal of the vacuum tube voltmeter specified by 4.7a to the positive (+) jack P6 and the negative (-) terminal of the voltmeter to the negative (-) jack P7. Both jacks arc located in the chassis.

c. Set the ON-OFF switch at ON. The voltmeter shall indicate 510 ± 10.

d. Set the ON-OFF switch at OFF.

e. Remove the voltmeter.

4.8.2.2 Operation of timing delay circuits. The tests of 4.8.221 and 4.8-222 shall be performed at a temperature between 65°F and 95°F. After the timing delay circuits have been calibrated in accordance with 3.5.2 the panel-chassis assembly shall be removed from the case, energized and allowed to warm up for 15 minutes before beginning the tests.

4.8.2.2.1 Five-second timing delay circuit. Procedures 4.8.2.2.1a through 4.8.2.2.1j shall be performed in sequence.

a. Set the ON-OFF switch at ON.

b. Set the TEST SELECTOR switch at 1.

c. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

d. Set the ON-OFF switch at OFF.

e. Connect a jumper across the VOLTAGE and LEAKAGE meter terminals. This jumper must be as short as possible and make positive electric contacts with both meter terminals.

f. Connect the electric timer specified by 4.7b to pins B and C of the 3-pin Adapter Cables connector.

g. Set the ON-OFF switch at ON.

h. Set the TEST SELECTOR switch at position 2.

i. Momentarily depress the START switch The timer shall run for 5 ± 0.25 seconds.

j. Disconnect the electric timer from the 3-pin ADAPTER CABLE connector.

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4.8.2.2.2 One-second timing delay circuit The procedures of 4.8.2.2.2a, through 4.8.2.2.2d shall be conducted in sequence immediately following the test of 4.8221.

a Set the TEST SELECTOR switch at 5

b. Connect the electric timer specified by 4.7b to the positive (+) binding post BP1, and the negative (-) binding post BP2 Both binding posts are located on the panel.

c. Depress and hold the START switch until the timer stops. The timer shall run for 1.0 ± 0.1 second.

d. Disconnect the timer and remove the jumper from the VOLTAGE and LEAKAGE meter. (This procedure may be omitted when the test of 4.826.4 immediately follows 4.8.2.2c).

4.8.2.3 Battery voltage test. This test shall be performed at a temperature between 65°F and 95°F after the test set has been calibrated in accordance with 3.5.3. The test set shall be energized and allowed to warm up for 15 minutes. Procedures 4.8.2.3a through 4.8.2.3k shall be performed in sequence.

a. Set the TEST SELECTOR switch at 1.

b. Momentarily depress the START switch. The test set shall start to cycle.

c. When panel lamp PL 11 lights, immediately set the ON-OFF switch at OFF.

d. Connect the positive terminal of a 6-volt heavy duty dry cell to pin V of the 19-pin ADAPTER CABLES connector and connect the negative terminal of the dry cell in series with the O. 1-megohm resistor specified by 4.7e to pin N of the 19-pin connector.

e Set the NO LOAD-LOAD switch at NO LOAD.

f. Set the ON-OFF switch at ON. The VOLTAGE and LEAKAGE meter shall indicate 60 ± 15 until the test set completes stepping and the HOME lamp lights.

g. Remove the dry cell and resistor from the 19-pin connector.

h. Connect the positive terminal of the dry cell to pin C of the 3-pin ADAPTER CABLES connector and connect the negative terminal of the dry cell in series with the O. 1-megohm resistor specified by 4.7e to pin B of the 3-pin connector.

i. Set the TEST SELECTOR switch at 2.

j. Momentarily depress the START switch. The VOLTAGE and LEAKAGE meter shall indicate 60 ± 15 for about 5 seconds.

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k. Remove the dry cell and the resistor from the 3-pin connector.

4.8.2.4 Battery insulation resistance test The tests of 4.8.2.4.1 and 4.8.2.4.2 shall be performed at a temperature between 65°F and 95°F after the VOLTAGE and LEAKAGE meter has been calibrated in accordance with 3.5.3. The panel-chassis assembly shall be removed from the ease, energized, and allowed to warm up for 15 minutes before beginning this test.

4.824.1 5 megohm. The procedures of 4.8.24.1a through 4.8.2.4.10 shall be performed in sequence:

a. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

b. Set the NO LOAD-LOAD switch at LOAD.

c. Set the TEST SELECTOR switch at 3.

d. If necessary, depress and hold the ADVANCE switch until the HOME lamp lights.

e. Set the 5 Meg-50 Meg switch at 5 Meg.

f. Connect the 5-megohm resistor specified by 4.7f from pin B of the TEST STAND connector to pin A of the 19-pin ADAPTER CABLES connector.

g. Turn the VTVM ADJ control until the Voltage and LEAKAGE meter indicates zero. If an indication of zero cannot be obtained by manipulating the VTVM ADJ control, the test set shall be regarded defective.

h. Momentarily depress the START switch. The HOME lamp shall be extinguished about 5 seconds after the START switch is depressed and Panel LAMP PL 1 and no other lamp shall light. The VOLTAGE and LEAKAGE meter shall indicate 46 ± 1 scale divisions.

i. Perform procedures 4.8.2.4.1i (1) through 4.8.24. Ii (3) for each step listed by table IL

CAUTION

To avoid possible harm from high voltages, 4.8.24.1i (l), 4.8.24.1i (2),
and 4.8.2.4.1 i (3) must be performed in sequenee.

(1) Immediately set the TEST SELECTOR switch at 4.

(2) With one end of the resistor remaining connected to pin B of the TEST STAND connector, connect the other end of the resistor to the pin indicated for the corresponding steps of table II.

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(3) Set the TEST SELECTOR switch at 3. In about 5 seconds the panel lamp indicated by the previous step (P11 for Step 1) shall be extinguished and the panel lamp listed for the ax-responding step and no other lamp shall light- The VOLTAGE and LEAKAGE meter shall indicate 46 ± 1 scale divisions.

j. Depress and hold the ADVANCE switch until the HOME lamp lights.

k. Disconnect the 5-megohm resistor from pin N of the 19-pin connector and connect it to pin C of the 3-pin ADAPTER CABLES connector.

L Set the TEST SELECTOR switch at 4.

m. Momentarily depress the START switch. The VOLTAGE and LEAKAGE meter shall indicate 46 ± 1 scale divisions.

n. Set the TEST SELECTOR switch at 3.

0. Disconnect the 5-megohm resistor.

TABLE II. Panel lamp connection.s.

step	Panel Lamp	Adapter Cable Connector	Pin
1	2	19-pin	B
2	3	19-pin	C
3	4	19-pin	D
4	5	19-pin	E
5	6	19-pin	F
6	7	19-pin	G
7	8	14-pin	A
8	9	14-pin	B
9	10	14-pin	C
10	11	19-pin	N

4.8.2.42 50 megohms. Immediately following the test of 4.8.2.4.1, procedures 4.8.2.4.2a through 4.8.2.4.2f shall be performed in sequence.

a. Set the 5 Meg-50 Meg switch at 50 Meg.

b. Connect the 50-megohm resistor specified by 4.7g between pin B of the TEST STAND connector and pin C of the 3-pin ADAPTER CABLES connector

c. Set the TEST SELECTOR switch at 4.

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d. Turn the VTVM ADJ control until the Voltage and Leakage meter indicates zero-

e. Momentarily depress the START switch. The VOLTAGE and LEAKAGE meter shall indicate 46 to 50.

f. Disconnect the 50-megohm resistor.

4.8.2.5 Battery short circuit test With the panel-chassis assembly removed from the ease and energized, perform procedures 4.8.2.5a through 4.8.2.5h in sequence.

a. If necessary, depress and hold the ADVANCE switch until the HOME lamp lights.

b. Set the TEST SELECTOR switch at 5.

c. Connect a 20 Amp. D.C. source in series with the 100 ampere D.C. ammeter specified by 4.7h between the positive (+) binding post (BP1) and the negative (-) binding post (BP2) observing polarity. Both binding posts are located. on the panel.

d. Set the SHORT CIRCUIT CURRENT meter to indicate 0 by means of the zero adjustment of the meter.

e. Remove the 6J5 tube.

f. Depress the START switch holding it only long enough to observe the readings of both the SHORT CIRCUIT CURRENT meter and the external ammeter. The readings of the two meters shall agree within 2 amperes.

g. Remove the dry cell and external ammeter.

h. Replace the 6.75 tube.

4.8.2.6 Continuity. The precision ohmmeter specified by 4.7i shall be used to make the resistance measurements required by 4.8.2.6.4 and 4.8.2.6.5. A resistance bridge of equal accuracy may be substituted for the ohmmeter when making the resistance measurements required by 4.8.26.1, 4.8.2.6.2 and 4.8.6.23.

4.8.2.6.1 Cable assembly CA-802. Measure the resistance between each pair of points listed by table III (see 6.5).

4.8.2.62 Pin V of ADAPTER CABLES connector PI to pin N of ADAPTER CABLES connector P2. With the test set not energized, measure the resistance between pin V of the 19-pin ADAPTER CABLE connector and pin N of the 14-pin ADAPTER CABLES connector. This resistance shall not exceed 0.15 ohm.

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4.8.2.6.3 Test circuits continuity. With the panel-chassis assembly removed from the case, and not energized, measure the resistance between pin B of the TEST STAND connector and the chassis. This resistance shall not exceed 0.15 ohm.

TABLE ILL Resistance test points.

From Socket of Connector PI	To
A	Lug P4
B	Lug P2
B	Contact Assembly 553712

Note: Resistance between each pair of points shall not exceed 0.15 ohm.

4.8.2.6.4 Continuity of VOLTAGE and LEAKAGE meter circuit, meter bypassed. With the panel-chassis assembly removal from the case, procedures 4.8.2.6.4a through 4.8.2.6.4m shall be performed in sequence.

a. Set the ON-OFF switch at OFF.

b. Connect a jumper across the VOLTAGE and LEAKAGE meter terminals. This jumper must be as short as possible and make positive electric contacts with both meter terminals.

c. Connect the precision ohmmeter between pins A and B of the 3-pin ADAPTER CABLES connector.

d. Set the ON-OFF switch at ON.

e. Set the TEST SELECTOR switch at 2 and the NO LOAD-LOAD switch at LOAD.

f. Momentarily depress the START switch- The ohmmeter shall indicate 0.15 ohm or less for about 5 seconds after the START switch is depressed.

g. Set the ON-OFF switch at OFF.

h. Remove the ohmmeter lead from pin A and connect it to pin C of the 3-pin ADAPTER CABLES connector.

i. Set the ON-OFF switch at ON.

j. Momentarily depress the START switch. The ohmmeter shall indicate 0.15 ohm or less for about 5 seconds after the START switch is depressed.

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k. Set the ON-OFF switch at OFF.

1. Remove the ohmmeter.

m. Remove the jumper from the terminals of the VOLTAGE and LEAKAGE meter.

4.8.2.6.5 Continuity through the self controlled rotary switch REL-X With the test set energized, procedures 4.8.2.6.5a through 4.8.2.6.5g shall be performed in sequence.

a. Set the TEST SELECTOR switch at 1.

b. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

c. Momentarily depress the START switch. The test set shall start to cycle, the HOME lamp shall be extinguished and panel lamp PL 1 and no other lamp shall light.

d. When lamp PL 1 lights, immediately set the ON-OFF switch at OFF.

e. Connect one lead of the precision ohmmeter to pin V of the 19-pin ADAPTER CABLES assembly.

f. Perform procedures 4.8.2.6.5f (1) through 4.8.2.6.5f (7) for each step of table IV.

(1) Set the NO LOAD-LOAD switch at NO-LOAD.

(2) Connect the free lead of the ohmmeter to the pin listed for the corresponding step by table IV under the column headed NO LOAD. The ohmmeter shall indicate 0.5 ohm or less.

(3) Set the NO-LOAD-LOAD switch at LOAD.

(4) Transfer the lead not connected to the pin listed for the corresponding step by table IV under the column headed LOAD. The ohmmeter shall indicate 0.5 ohm or less.

(5) Disconnect the connection made under (4) above.

(6) Set the ON-OFF switch to ON. The test set shall cycle and the panel lamp listed by table IV for the corresponding step shall be extinguished The panel lamp listed for the next step (or, for Step II only, the HOME lamp) and only that lamp shall light.

(7) When the panel lamp for the next step lights, immediately set the ON-OFF witch at OFF.

g. Disconnect the ohmmeter.

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TABLE IV. No-load switch position.

		No-load Adapter Cable		Load Adapter Cable	
Step	Lamp	Connector	Pin	Connector	Pin
1	1	19-pin	A	19-pin	H
2	2	19-pin	B	19-pin	J
3	3	19-pin	c	19-pin	K
4	4	19-pin	D	19-pin	L
5	5	19-pin	E	14pin	D
6	6	19-pin	F	14-pin	E
7	7	19-pin	G	14-pin	F
8	8	14-pin	A	14-pin	G
9	9	14-pin	B	14-pin	J
10	10	14-pin	c	14-pin	K
11	11	19-pin	N	19-pin	M

4.8.2.6.6 D.C resistances and continuity through ON-OFF switch.

4.826.6.1 VOLTAGE and LEAKAGE meter coil resistance With the panel-chassis assembly removed from the case and Cable Assembly CA-731 disconnected from the panel chassis assembly, procedures 4.8.2.6.6.1a through 4.8.2.6.6.1j shall be performed in sequence at a temperature of 65°F to 95°F.

a. Set both the ON-OFF switch and the TEST SELECTOR switch at OFF.

b. Connect a 6-volt heavy duty dry cell, the 50,000-ohm resistor specified by 4.7j, and the 50,000-ohm potentiometer specified by 4.7k in series, connecting the center tap of the potentiometer either to the battery or the resistor.

e. Observing polarity, connect the series combination of b, above, across the terminals of the VOLTAGE and LEAKAGE meter.

d. Turn the potentiometer shaft to the approximate center position.

e. Set the ON-OFF switch at ON.

f. Adjust the potentiometer for full scale reading cm the VOLTAGE and LEAKAGE meter.

g. Set the ON-OFF switch at OFF.

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h. Connect the decade resistor specified by 4.71 across the terminals of the VOLTAGE and LEAKAGE meter.

i. Set the ON-OFF switch at ON.

j. Adjust the decade resistor until the VOLTAGE and LEAKAGE meter indicates exactly 50. The decade resistor shall indicate 1,000 ± 10 ohms.

4.8.2 .6.6.2 Continuity of the ON-OFF switch with the switch at the OFF position. Procedures 4.8.2.6.6.2a and 4.8.2.6.6.2b shall be performed immediately following 4.826.6.1j.

a. Set the ON-OFF switch at OFF. The VOLTAGE and LEAKAGE meter shall indicate O.

b. Disconnect the dry cell, resistor and potentiometer from the terminals of the VOLTAGE and LEAKAGE meter.

4.8.2.6.63 Resistor R14. With the panel-chassis assembly removed from the case, procedures 4.8.2.6.6.3a, 4.8.2.6.6.3b and 4.8.2.6.6.3c shall be performed in squence.

a. Set both the ON-OFF switch and the TEST SELECTOR switch at OFF.

b. Set the 5 Meg-50 Meg switch at 50 Meg.

c. Connect the ohmmeter or resistance bridge specified by 4.7m to the chassis and the negative jack P7, located in the chassis. The ohmmeter or resistance bridge shall indicate 0.1 megohm ± 1 percent

4.8.2.6.6.4 Resistor R5. Immediately after performing 4.8.2.6.6.3c, set the 5 Meg-50 Meg switch at 5 Meg. The ohmmeter shall indicate 10,000 ohms ± 1 percent. Remove ohmmeter.

4.%2.7 Insulation resistance.

4.8.2.7.1 Conditioning. To expedite production, insulation resistance may be measured at any temperature above 70°F and at any relative humidity above 70 percent. When a test set fails to meet requirements under these conditions, it shall then be conditioned for at least 24 hours at a temperature of 77 ± 2°F and a relative humidity of 75 ± 5 percent. Within one-half hour after such conditioning, procedures 4.8.2.7.2a through 4.8.2.7.2h shall be performed at a relative humidity of 60 percent or more.

4.8.2.7.2 Test procedures. Procedures 4.8.2.7.2a through 4.8.2.7.2f shall be performed in sequence. Following 4.8.2.7.2f all measurements required by 4.8.2.7.2g and 4.8.2.7.2h shall be made. The resistance device specified by 4.7n shall be used to determine all insulation resistance.

a. With the test set connected to a 115 volt A.C. supply, set the ON-OFF switch at ON.

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b. If the HOME lamp does not light, depress and hold the ADVANCE switch until the HOME lamp lights. Set the test selection to Position 1.

c. Disconnect Cable Assembly CA-731 from the test set.

d. Remove the panel-chassis assembly from the case

e. Set the TEST SELECTOR switch at OFF.

f. Set the NO LOAD-LOAD switch at NO LOAD.

g. Measure the insulation resistance between all pairs of pins listed by table V. Each resistance measured shall be 500 megohms or more.

h. Measure the insulation resistance between all pairs of pins listed by table VI. Each resistance measured shall be 50 megohms or more.

TABLE V. 500-Megohm insulation resistance test pin pairs.

19-Pin Connector PI

A to V	A to B	B to D	C to G	A to GRD
B to V	A to C	B to E	D to E	B to GRD
C to V	A to D	B to F	D to F	C to GRD
D to V	A to E	B to G	D to G	D to GRD
E to V	A to F	C to D	E to F	E to GRD
F to V	A to G	C to E	E to G	F to GRD
G to V	B to C	C to F	F to G	G to GRD
				V to GRD

3-Pin Connector P3

C to A
C to B
C to GRD

Wiper REL-X4 to GRD
Moving contact REL-1D to GRD

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TABLE VL 50-Megohm insulation resistance test pin pairs.

19-Pin Connector PI

H to J	J to K	K to M	L to R	M to GRD	P to GRD
H to K	J to L	K to N	L to S	N to P	R to S
H to L	J to M	K to P	L to T	N to R	R to T
H to M	J to N	K to R	L to U	N to S	R to U
H to N	J to P	K to S	L to GRD	N to T	R to GRD
H to P	J to R	K to T	M to N	N to U	S to T
H to R	J to S	K to U	M to P	N to GRD	S to U
H to S	J to T	K to GRD	M to R	P to R	S to GRD
H to T	J to U	Lto M	M to S	P to S	T to U
H to U	J to GRD	L to N	M to T	P to T	T to GRD
H to GRD	K to L	L to P	M to U	P to U	

14-Pin Connector P2

A to B	B to D	c to G	D to M	F to L	K to L
A to C	B to E	C to J	D to GRD	F to M	K to M
A to D	B to F	C to K	E to F	F to GRD	K to GRD
A to E	B to G	C to L	E to G	G to J	L to M
A to F	B to J	C to M	E to J	G to K	L to GRD
A to G	B to K	C to GRD	E to K	G to L	M to GRD
A to J	B to L	D to E	E to L	G to M	
A to K	B to M	D to F	E to M	G to GRD	
A to L	B to GRD	D to G	E to GRD	J to K	
A to M	C to D	D of J	F to G	J to L	
A to GRD	C to E	D to K	F to J	J to M	
B to C	C to F	D to L	F to K	J to GRD	

3-Pin CONNECTOR P3

Ato B
A to GND
B to GND

TEST STAND CONNECTOR P4

A to GND

Wiper REL-X5 to GND
Moving contact REL-1 C to GND

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4.8.2.7.3 Cable assembly CA-802 (see 6.5). At a temperature between 65°F and 95°F and a relative humidity between 45 and 95 percent, measure the insulation resistance between pins A and B of the connector of Cable Assembly CA-802 using the insulation measuring device specified by 4.7n. Test potential shall not be applied for more than 5 seconds. The resistance measured shall be 50 megohms or more.

4.8.2.8 Continuity test switch operation. With the test set energized, procedures 4.8.2.8a through 4.8.28d shall be performed in sequence.

a. Set the TEST SELECTOR switch at 1.

b. If the HOME lamp is not lit, depress and hold the ADVANCE switch until the HOME lamp lights.

c. Progressively connect and remove a jumper between the points listed by table VII, turning the CONTINUITY TEST switch to the position indicated by the table. The green panel lamp PL 14 shall light or be extinguished as indicated by table VII. Panel lamp PL 14 shall not be lit in absence of a jumper.

d. Remove the jumper. Panel lamp PL 14 shall be extinguished.

TABLE VII. Location of jumper.

Step	Position of Continuity Test Switch	Adapter Cables Connector	Across Pins	Panel Lamp P14 Lit or Not Lit
1	1	19-pin, PI	P and S	Lit
2	2	19-pin, PI	P and S	Not Lit
3	2	19-pin, PI	R and T	Lit
4	3	19-pin, P1	R and T	Not Lit
5	3	14-pin, P2	Land M	Lit

4.8.2.9 Test circuits operation. With the test set energized, procedures 4.8.2.9a through 4.8.2.9f shall be performed in sequence.

a. Set the TEST SELECTOR switch in position 1.

b. If the HOME lamp is not lit, depress and hold the ADVANCE SWITCH until the HOME lamp lights.

c. Connect a jumper across pins A and B of the TEST STAND connector.

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d. Depress the TEST switch. The green panel lamp PL 12 adjacent to the TEST switch shall light.

e. Release the TEST switch. Panel Lamp PL 12 shall be extinguished

f. Remove the jumper.

4.8.2.10 Operation of self controlled rotary switch, REL-X With the test set energized, procedures. 4.8.2.10a through 4.8.2.10d shall be performed in sequence.

a Set the TEST SELECTOR switch at 1.

b. Momentarily depress the START switch. The test set shall cycle until the HOME lamp lights. (This procedure may be omitted if the HOME lamp is already lit).

c. Momentarily depress the START switch. When panel lamp PL 1 lights, immediately depress and hold the ADVANCE switch. The test set shall cycle rapidly until the HOME lamp lights.

d. Release the ADVANCE switch.

4.9 Durability tests.

4.9.1 Repeated package impact. For this test, the test set in its shipping container shall be placed on a vibration platform equipped with barriers so as to allow free movement of the test set relative to the platform but prevent it from falling off the platform. The motion of the platform shall be such that any point on the platform moves in a circular path one inch in diameter with the plane of the circle perpendicular to the platform. The frequency shall be within the range of 240 to 300 cycles per minute. The vibration shall be for one-half hour on each face of the shipping container.

4.9.2 Shock, transportation, storage and handling. The test set in the shipping container shall be dropped cornerwise from a height of 30 inches onto a solid steel or concrete surface in such a manner that one corner and one face of the shipping container absorbs the full force of the fall. (The height of 30 inches refers to the distance from the steel or concrete surface to the nearest comer of the test set when suspended prior to fall). The fall shall be freefall in that no ropes or other suspended media are attached to the container during the fall. Damage to the shipping container which could result in loss of contents or failure of the container in subsequent handling shall result in the packaging and packing being regarded defective and in the ease of a periodic test sample may require that the contractor repackage and repack the lot represented by the sample. Minor container damage such as chipping of wood members, dents and/or paint chipping is not evidence that the shipping container has failed this test. After the shipping container has been examined, the test set shall be removed from its shipping container and unit container and examined. A test set shall be regarded as defective when there is visual evidence of mechanical damage or the hardware has loosened.

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4.9.3 Vibration, transportation storage and handling. The test set shall be rigidly fastened to the vibration table by means of a suitable fixture. Simple harmonic excitation shall be applied parallel to each of the three principal axes of the device for equal periods of time. The frequency range from 10 to 60 cps may be covered either by slowly cycling at a logarithmic rate between the limits, or by at least 24 discrete frequency steps which have a logarithmic distribution. The vibration amplitude shall be maintained at 2 ± 0.2g (zero-to peak) or 0.10 ± 0.01 inch (peak to peak) whichever is less. The duration of the test in each direction is 4 hours. The total test duration shall be 12 hours. There shall be no visible damage or loosening of hardware as a result of the vibration.

4.10 Inspection of packaging. Packaging shall be visually inspected for conformane to section 5.

5. PACKAGING.

5.1 Preservation packaging. Presrvation packaging shall be level A or C as specified in the contract (see 6.2).

5.1.1 Level A.

5.1.1.1 Preparation of cable assemblies. Prior to unit packaging of the test set, Cable Assemblies CA-731 and CA-802 shall be individually packaged in accordance with LD 284866 and placed in the cable well of the test set.

5.1.1.2 Unit container. Each test set shall be packaged as specified by 5.1.1.2a, 5.1.1.2b, and 5.1.1.2C.

a Each test set, with cable assemblies placed in the cable well, shall be packaged in a corrugated fiberboard box conforming to PPP-B-636, Type I (CF), Class 2, Grade 5 (W5C), Style RSC, “B” flute, bursting strength 275 lb minimum. Sufficient corrugated fiberboard sheet shall be used within the fiberboard box to prevent direct contact between protrusions of the test set ease and the walls of the fiberboard box Fiberboard sheet shall conform to PPP-F-320, Type CF, Class-weather resistant, Variety SW, Grade W5C, “B” flute.

b. Each test set, prepared as specified by 5.1.1.2a, shall be enclosed in a close fitting heat sealed bag fabricated from watervaporproof, flexible, heat sealable barrier material cpmforming to MIL-B-131, Class I.

c. Each test set, prepared as specified by 5.1.l.2.a and 5.1.1.2b shall be over packaged in a fiberboard box conforming to PPP-B-636, Type optional, Class 2, Grade 5, Style RSC, bursting strength 275 lb minimum. The outer container shall be closed by adhesive, tape or other practicable means which will not permit the possibility of damage to the barrier material.

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5.12 Level C. Each test set shall be individually packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A or C as specified in the contract or order (see 6.2).

5.21 Level A. Each unit container, packaged as specified by 5.1.12 shall be packed either in a wood, cleated-plywood box conforming to PPP-B-601, overseas type, Style A or B, average load or a nailed wood box conforming to PPP-B-621, Class 2, Style 2, 2-1/2,3 or 4, bad-type 2, with the lumber surfaced on both sides. When a nailed wooden box is used, lumber shall conform to MIL-STD-731, Group I, Group II, Groups I and II or Group III.

5.2.2 Level C. Unit containers, packaged as specified by 5.1.2 shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers a shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), all marking shall be in accordance with MIL-STD-1 29. The following marking shall appear on the top, one end, and one side of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The Test Set Mark 127 Mod 4 is used with interconnecting cable assemblies and test adapters from Accessory Set, Mark 1 Mod 1 to perform tests on batteries for use in underwater ordnance equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 21.1 and 21-2).

c. Whether a first article inspection is required (see 3.1 and 4.4).

d. Any data required (see 3.2).

e. The agency that is to perform the inspection (see 4.1).

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f. Inspection conditions, if' other than as specified (see 43).

g. Production lot size and sample size, if other than as specified (see 4.6).

h. Authority for classification of unlisted characteristics (see 4.8.1).

i. When first article samples are waived because of having been submitted on a previous recent contract. the purchase document should contain a statement that the standards of workmanship exhibited by the previously submitted first article sample shall state the minimum requirements in the current contract order, relative to workmanship (see 3.8).

j. Level of packaging required (see 5.1).

63 First article When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting offficer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending mntract-

6.4 Definitions. The following definitions are used throughout this specification and are herein defined for clarity.

6.4.1 Test Set. "Test set" means Test Set Mark 127 Mod 4.

6.4.2 Panel-chassis assembly. 'Panel-chassis assembly" means Panel-chassis Assembly, Drawing 1362835.

6.43 Case "Case" means Instrument Case (Metal), LD276015.

6.4.4 Panel. "Panel" means that portion of the panel-chassis assembly that is accessible with the panel-chassis assembly installed in the case and the lid of the case removed

6.4.5 Chassis. "Chassis" means that portion of the panel-chassis assembly that is inaccessible with the panel-c hassis assembly installed in the case and the lid of the case removed.

6.4.6 Pilot lamp or panel lamp. "Pilot lamp" (See Drawing 1362891) and "panel lamp" are used interchangeably.

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6.4.7 Energized. A test set is ‘energized” when it is connected to a 115 volt A.C. source by means of Cable Assembly CA-731 and the ON-OFF switch is at the ON position.

6.5 Electrical connections. Test probes shall not be used to make electrical connections to the connector of Cable Assembly CA-802 Jigs made up of mating connectors shall be used. Mating connectors can be wired or their solder pots can be used as test points.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-21369C(OS)	2. DOCUMENT DATE (YYMMDD) 23 July 1990

3. DOCUMENT TITLE
Test Set, Mark 127 Mod 4

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commanding Officer Naval Ordnance Station	b. TELEPHONE (Include Area Code) (1) Commercial (301)743-4510 (2) AUTOVON 364-4510
c. ADDRESS (Include Zip Code) Code 3990 Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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